CHENMING GE

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RESEARCH INTERESTS

Human-Computer Interaction (HCI), Generative UI, Computer Use Agent, Human-AI Interaction

EDUCATION

University of Michigan, Ann Arbor — Ann Arbor, USA
B.S.E. in Computer Science — *Aug 2025 - May 2027 (Expected)*

- **GPA:** 3.86/4.00

Shanghai Jiao Tong University — Shanghai, China
B.Eng. in Electronic and Computer Engineering — *Aug 2023 - Aug 2027 (Expected)*

PUBLICATIONS

A11yLTLNav: Automatic Detection of Accessibility Navigation Failures
Ge, C., Peng, K., Shi, C., Greenman, B., Jiang, Y.
arXiv:2609.17959. Submitted to the ACM CHI Conference on Human Factors in Computing Systems (CHI 2027).

Revision-Based Learning from Interactive Experience and Feedback
Li, J., **Ge, C.**, Gao, A., Vempati, R., Peng, Y.-H., Wu, J.
Submitted to the 2027 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL 2027).

RESEARCH EXPERIENCE

Automatic Detection of Accessibility Navigation Failures — May 2026 – Sep 2026
Research Assistant at University of Utah — *Advisor: Dr. Yue Jiang and Dr. Ben Greenman*

- Developed **A11yLTLNav**, a property-based checker that checks **accessibility navigation failures** encountered by Blind and Low-Vision (BLV) users when interacting with generated UIs..
- Formalize common accessibility failures from previous user studies into **Linear Temporal Logic (LTL)** formulas and test them by randomly generating interaction traces on websites.
- Implemented an **web agent** baseline that navigates websites through screen-reader feedback and keyboard-only interactions to detect accessibility failures.
- Evaluated on generated websites, achieving **88.7% precision**, outperforming baselines by **31.4%**.

Revision-based Learning from Interactive Experience and Feedback — May 2026 – Oct 2026
Research Assistant at Purdue University — *Advisor: Dr. Jason Wu*

- Investigating how user **interaction** traces can provide implicit feedback for improving models' ability to generate **generative UIs** with better **usability**.
- Built a pipeline to generate websites, collect task interaction traces from user study, diagnose usability critiques from the traces, and automatically repair websites with critiques.
- Explored using websites revised with usability critiques to **fine-tune** models for generating UIs with better usability.
- Explore baselines using trained **Computer Use Agents** simulating users' behavior and coding agents to give feedback of usability of generated websites.

Adversarial Co-Evolution of Reflection Rules and Evolving Website for Web Agents Mar 2026 – Jun 2026
Research Assistant at University of Michigan — *Advisor: Prof. Honglak Lee*

- Built **WebCoEvo**, an **Adversarial Co-Evolution** framework for web agents when websites' version updating, pairing a dynamic UI version change generator using **coding agents** to update its version with a **web agent**.
- Extract generalizable **meta-level reflection rules** from failure modes of web agents, outperforming **ExpeL** baselines by **19.6%**.
- Implementing a **Knowledge Graph** based pipeline to automatically identify version-affected tasks, replacing manual annotation for evolving websites.
- Containerized multi-version web apps via **Docker** on BrowserGym and AgentLab.

SKILLS

Frameworks	PyTorch, Hugging Face, CUDA, Docker, Linux, React, Vue, Vite, OpenAI SDK
Languages	Python, C/C++, Java, Rust, HTML/CSS/JavaScript, TypeScript, SQL, MATLAB
Tools	Figma, Illustrator, Remotion, MATLAB, Codex, Claude Code, SolidWorks, LabVIEW